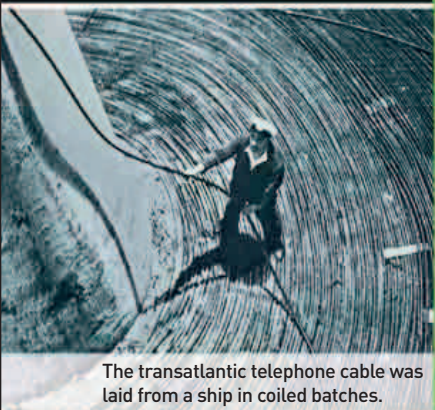


“... THE FIRST ARTIFICIAL EARTH SATELLITE HAS BEEN BUILT.”

Telegraph Agency of the Soviet Union (TASS), published in *Pravda* newspaper, October 5, 1957



The transatlantic telephone cable was laid from a ship in coiled batches.

fishing waters from a chemical factory. The mercury had accumulated in fish and shellfish, and had poisoned the people who ate them. This incident was one of the first well-documented cases of the effects of toxic chemicals polluting food chains.

In July, British chemist **Dorothy Hodgkin** (see 1945) published a study of the **structure of vitamin B12**—a vitamin needed to prevent pernicious anemia. Using X-ray crystallography, she found that the vitamin contained a ringlike structure, porphyrin, that surrounded a central atom of the element cobalt.

The **world's first underwater telephone cable** became operational on September 25. It ran between the US and Europe under the Atlantic Ocean. In the first 24 hours of operation, the service hosted 588 London to US telephone calls, which were clearer than any previous transatlantic communication.

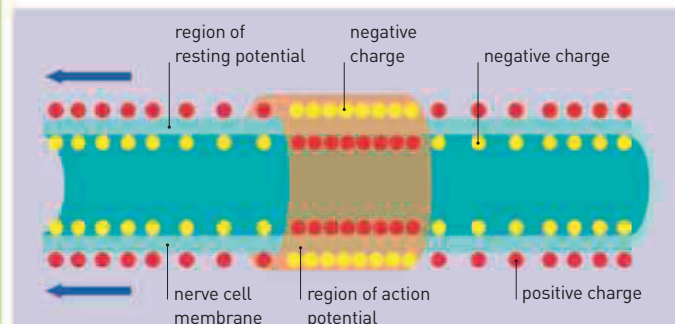
In September, IBM launched the 305 RAMAC (Random Access Memory Accounting Machine)—the **first computer with a hard disk drive** and random access memory. RAMAC weighed over a ton and stored 5MB of data on a stack of 50 large disks.

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THE NUMBER OF DAYS THAT SPUTNIK 1 TRANSMITTED SIGNALS BACK TO EARTH

THE HAMILTON ELECTRIC 500, launched in January, was the **first ever wristwatch that did not need winding**. Although its battery life was relatively short, the innovation proved to be very popular.

Danish chemist **Jens Skou** (b.1918) published the basis of how the **nervous system** works. He discovered molecules in the cell membranes of crab nerves that pump ions to “charge up” membrane surfaces, using up cellular energy in the process. This makes the molecules excitable so that they can carry



CREATING A NERVE IMPULSE

Membranes around nerve fibres are electrically charged because they contain protein “pumps” that push ions (charged particles) of sodium out and pull potassium ions in. This makes positive charge accumulate on the outer surface: a so-called resting potential. When stimulated, protein channels open in the membrane, making positive charge leak in. This reverses the resting potential to create an action potential. The region of action potential fires down the nerve fiber membrane as a nerve impulse.

an impulse (see panel, below) when stimulated.

On October 4, Russia launched Sputnik 1—the first artificial satellite to orbit Earth.

At the start of the nuclear arms race, and prompted by US President Eisenhower’s “atoms for peace” speech in 1953, the United Nations set up the **International Atomic Energy Agency (IAEA)** in July 1957 to control and develop **atomic energy**. Thirteen years later, the IAEA would oversee the Treaty on Non-Proliferation of Nuclear Weapons.

In the same year, aviation company **Boeing** launched the **first commercial jet airliner**, the 707. It marked the dawn of a new age of air travel powered by turbine engines that made aeroplanes fly higher and faster than before.

In November, American physicist **Gordon Gould** (1920–2005) suggested a method of amplifying light into an intense beam and coined the term for it as **laser** (Light Amplification by Stimulated Emission of Radiation). An operational laser, however, would not be produced until 1960.

British scientist **James Lovelock** (b.1919) invented the **electron capture detector (ECD)** as an ultrasensitive way of analyzing gas mixtures.

The device emits electrons (negatively charged particles), which are then absorbed by certain gases to produce a signal. It helped environmentalists detect tiny quantities of atmospheric pollutants, such as the pesticide DDT and chlorofluorocarbons (CFCs)—ozone-depleting (see 1973) compounds used in refrigeration and as propellants in aerosol cans.



James Lovelock holds ECD
The electron capture detector (ECD) picked up the tiniest amounts of electron-binding chemicals in the atmosphere, such as the chlorine in pesticides and other compounds.

September
IBM launches first computer with a hard disk drive

November 4
Minamata disease in Japan traced to heavy metal poisoning

January 3 Launch of first electric watch

February
Sodium-potassium pump in nerve cells discovered

July 29 International Atomic Energy Agency is established

October 4 Russia launches first artificial satellite, Sputnik 1

October 28 Commercial jet airliner Boeing 707 takes its first flight

November
American physicist Gordon Gould describes theory of lasers

ECD is invented to measure traces of chemicals in the atmosphere